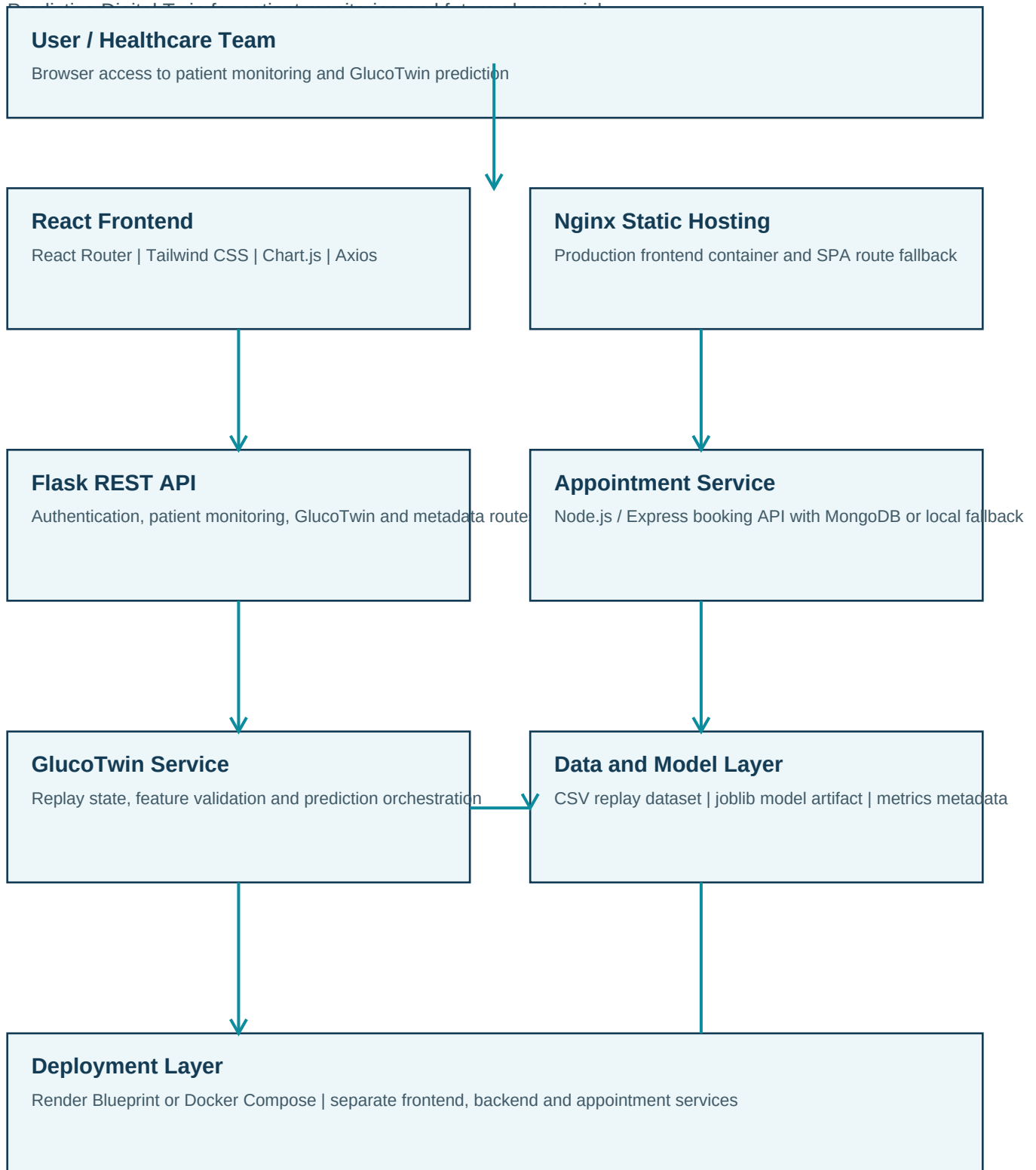


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# Gluco System Architecture



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# GlucoTwin Prediction Flow

## 1. Historical Profile

Patient ID, sex and HbA1c are loaded from the replay dataset



## 2. Simulated Dynamic Observation

Glucose, heart rate and IBI advance through the patient time series



## 3. Temporal Features

15 / 30 / 60 minute glucose change, slope and variability are calculated



## 4. Existing Model Inference

Saved scikit-learn Random Forest pipeline produces predict\_proba output



## 5. Future 2-Hour Risk

Risk of glucose rise  $\geq 30$  mg/dL within the next 2 hours



## 6. Explainable UI

Current state, trends, risk level and why risk is changing are displayed



## Data and Safety Boundary

The stream is simulated from open-source observations. The prototype does not claim real sensors, clinical validation or replacement of professional medical advice.